

Computer Vision and Image Processing

Final Project Report

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I. Title of the project:

Satellite based heat monitoring and vegetation analysis

II. Problem:

Input: Three landsat bands are used as input: red reflectance (SR_B4), near infrared reflectance (SR_B5) and land surface temperature (ST_B10). A quality assurance pixel mask is applied to filter out cloudy, shadowed regions and replace missing pixels before preprocessing. These bands are then merged into one image patch with 3 channels of size 64x64 pixels at 30m spatial resolution. These are extracted from scenes covering 6000×6000 pixels per city. Two geographic regions are used: Phoenix for model training and Las Vegas for testing the model on unseen data.

Output: A pixel level prediction map per geographical location which represents the spatial hotspots prediction through sliding window patches. This map shows the hotspot regions. Other outputs include the NDVI deficit estimates of the hotspot, the increase percentage of recommended vegetation and a hotspot comparison between the proposed deep learning model and baseline NDVI. Visualization of all findings include hotspot prediction overlay on satellite imagery and Grad-CAM activation maps to visualize which features the model focused on during training. All results are compiled into a structured PDF report.

III. Dataset:

1. Source: Earth Resources Observation and Science (EROS) Center. (2020). Landsat 8-9 Operational Land Imager / Thermal Infrared Sensor Level-2, Collection 2 [dataset]. U.S. Geological Survey. <https://doi.org/10.5066/P9OGBGM6>
2. Label generation was done using z-score local anomaly detection. A pixel was flagged as a hotspot if the corresponding land surface temperature(LST) value deviated significantly from its local neighbourhood with a threshold of 1.1.
3. Region A for training: Phoenix was selected because of its extreme urban heat island effect, low vegetation and near zero cloud cover during summer months. LST range: 11.16°C to 77.40°C, mean NDVI: 0.094, with 5.21% of valid pixels as hotspots.
4. Region B for testing: Las Vegas has a dry climate but distinct urban design. It was used as an unseen geographic test for model generalization.
5. Class balancing: From the available pixels, 15,000 each of hotspot and non hotspots patches were sampled. It was balanced to prevent model bias towards non hotspot classes.
6. Train validation split: 30,000 total patches were split in 80/20 ratio.

IV. Algorithm:

1. **QA masking and band conversion:** The QA_Pixel band encodes per pixel quality as bit flags. Pixels are marked as fill - bit 0, cloud - bit 3 and cloud shadow - 4. These bits are removed and set to NaN before processing. The remaining bands are converted to units using scale factors from the MTL metadata. The SR_B4 and SR_B5 bands are converted to surface reflectance and ST_B10 are converted to land surface temperature in Celsius.
2. **Label generation using Z-Score local anomaly detection:** Pixels are labelled as hotspot if their LST is much higher than their neighbouring pixels. A z-score is calculated for each pixel based on the mean and standard deviation. Pixels with a z-score over 1.1 are hotspots and other are non hotspot. This approach allows for better generalization across different cities and baseline climates.
3. **NDVI baseline for classical remote sensing:** NDVI is calculated as $(SR_B5 - SR_B4) / (SR_B5 + SR_B4)$. Pixels with low NDVI and high LST are flagged as hotspots. This serves as a remote sensing baseline to compare with the deep learning model.
4. **Patch extraction and normalization:** 64×64 patches are extracted centered on each labelled pixel across 3 channels (SR_B4, SR_B5, ST_B10). Edge pixels are skipped to avoid patches that might be out of bounds. Each patch is normalized to zero mean and unit variance inline during extraction. 15,000 hotspot and non hotspot patches are sampled to maintain class balance.
5. **Deep learning model, fine-tuned ResNet18 with custom head:** Pretrained ResNet18 is loaded with layer 1 and layer 2 frozen. Layer 3, layer 4 and custom head are kept trainable. The default classification head is replaced with a custom 3-layer FC network: FC (512→256) → BatchNorm → ReLU → Dropout(0.4), FC(256→64) → ReLU → Dropout(0.3), FC(64→1) → Sigmoid. This gives a hotspot probability between 0 and 1.
6. **Training:** The model is trained using Adam optimizer with a ReduceLROnPlateau scheduler and focal loss to handle difficult samples near hotspot boundaries. Online data augmentation is applied during training, such as random horizontal flips, vertical flips and 90° rotations. Early stopping is used.
7. **Grad-CAM:** Grad-CAM activation maps are generated after training to visualize which spatial regions within each patch influenced the model's prediction the most.

Code on my own: Z-score label generation, QA pixel masking and band unit conversion, NDVI hotspot baseline, patch extraction with class balancing, online data augmentation, custom ResNet18 classification head, focal loss implementation, Grad-CAM activation maps, CNN vs NDVI comparison maps and hotspot overlays on satellite imagery.

From online resources: Pretrained ResNet18 weights, PyTorch for model training, rasterio for reading landsat GeoTIFF files, numpy and scipy for array operations and local statistics, scikit-learn for evaluation metrics and ReportLab for PDF report generation.

V. Results:

Model Performance:

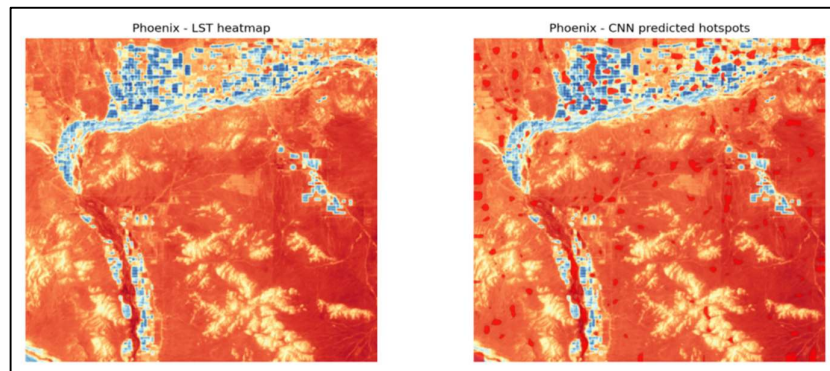
Metric	Phoenix(Validation set)	Las Vegas (Test set)
Accuracy	86.62%	86.25%
Precision	0.8631	0.8675
Recall	0.8703	0.8556
F1 Score	0.8667	0.8615

CNN vs NDVI comparison:

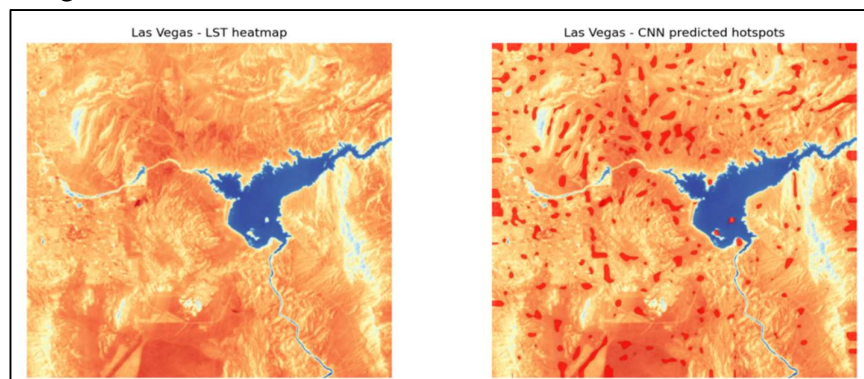
Flagged hotspots	Phoenix	Las Vegas
Both CNN and NDVI flagged	23,896 (1.06%)	45,933 (2.04%)
CNN only	82,119 (3.65%)	110,141 (4.90%)
NDVI only	23,768 (1.06%)	46,226 (2.05%)
Non Hotspots	2,120,217 (94.23%)	2,047,700 (91.01%)

Qualitative Results:

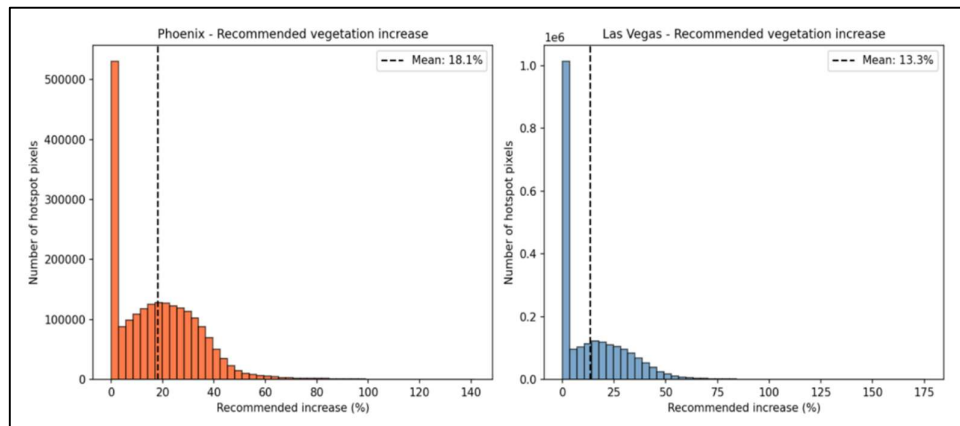
1. Phoenix hotspot overlay: LST heatmap and CNN predicted hotspots overlaid. Shows the spatial output of hotspot predicted by model.



2. Las Vegas hotspot overlay: LST heatmap and CNN predicted hotspots overlaid. Shows generalization of the model on unseen data.



- Vegetation gap histogram: Shows the recommended vegetation increase needed for both cities.



VI. Analysis:

Advantages:

- Z-score labelling detects hotspots relative to local neighbourhood, allowing generalization across cities with different baseline climates.
- Per patch normalization removes absolute temperature dependency, allowing the model to transfer across different datasets without retraining.
- Focal loss with class balancing prevents bias toward the dominant non-hotspot class
- Grad-CAM provides interpretability, showing that the model focuses on thermal boundaries and urban surfaces rather than noise.
- CNN captures thermally irregular regions that NDVI misses, flagging 3.65% in Phoenix and 4.90% in Las Vegas, adding value to the baseline classification.

Limitations:

- Z-score threshold of 1.1 was used for label generation. A stricter threshold would produce more confident hotspot labels.
- Training on a single Phoenix scene limits generalization to cities with significantly different urban layouts or climates
- Spatial prediction map is generated over a cropped 1500×1500 pixel region rather than the full scene due to computational constraints.

VII. Bibliography:

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3. <https://casoilresource.lawr.ucdavis.edu/software/grass-gis-raster-vector-and-imagery-analysis/raster-operations/working-landsat-data>
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